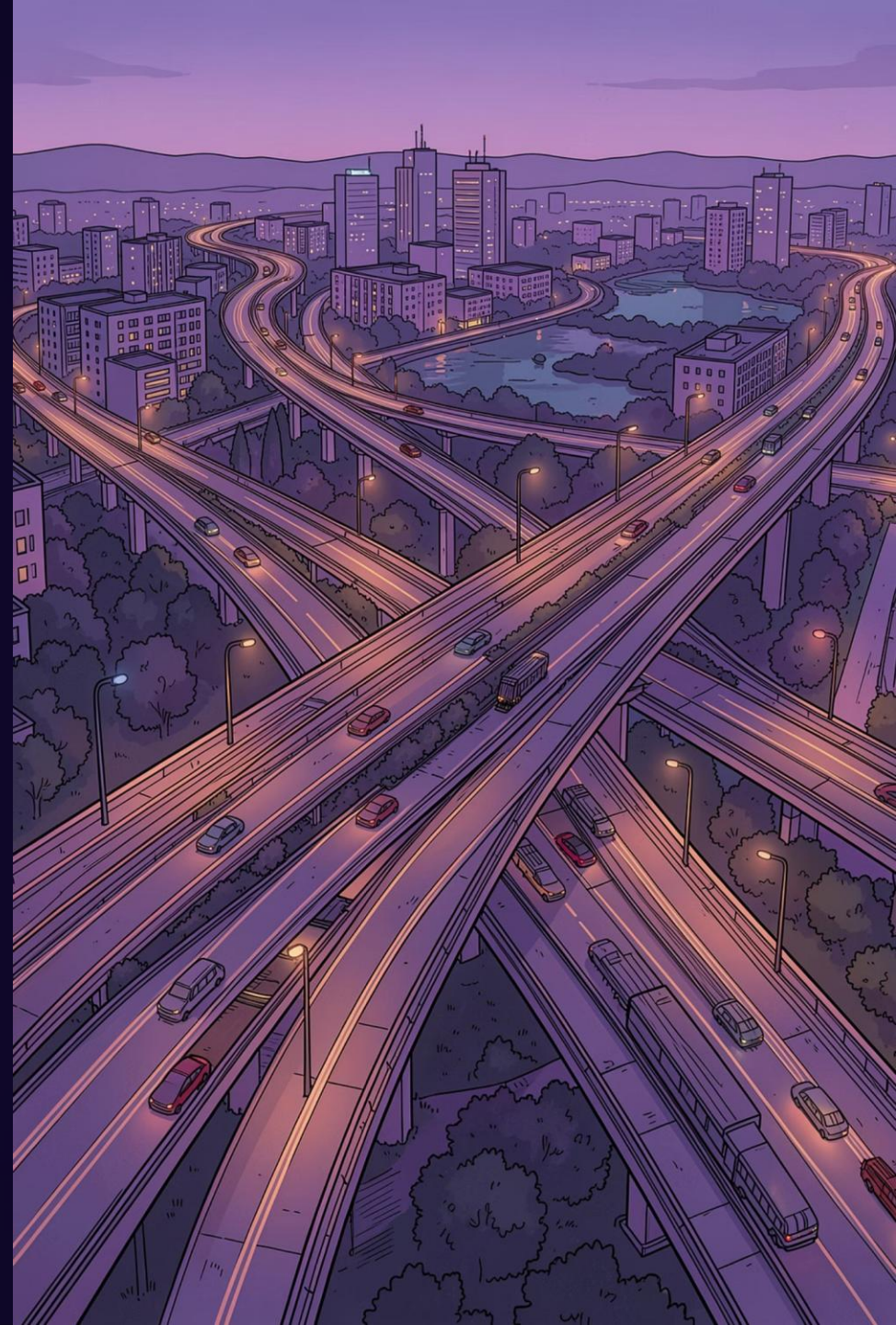


ROAD ACCIDENT ANALYSIS PROJECT



BUSINESS PROBLEM

- A national transport safety authority is facing a serious issue with increasing road accidents across different regions. However, they do not clearly understand:



- • Where accidents are happening most



- • When accidents are happening



- • What conditions lead to severe accidents



- • Which factors increase risk (weather, road type, time, etc.)

BUSINESS GOAL

- The goal of this analysis is to:

- **Identify key patterns, risk factors, and high-accident zones to help reduce road accidents and improve public safety policies.**



WHAT THE AUTHORITY WANTS TO ACHIEVE



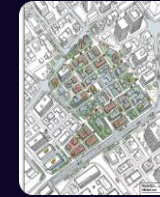
- • Reduce accident frequency in high-risk areas



- • Understand when accidents are most likely to occur



- • Identify conditions that lead to severe accidents



- • Improve road safety planning and enforcement



- • Support data-driven decision making for traffic control

BUSINESS QUESTIONS

- Overall Accident Impact

- ### • What is the total number of casualties recorded in the dataset, and how are they distributed across the data period?

- Severity Analysis

- ### • How are casualties distributed across different accident severity levels, and which severity level contributes the most to overall casualties?

- Vehicle Involvement Analysis

- ### • Which vehicle types are most frequently involved in accidents resulting in casualties?

- Road Type Risk Analysis

• Which types of roads are associated with the highest number of casualties, and what does this indicate about road safety conditions?

- Geographic Risk Analysis

• Which districts or regions report the highest number of casualties, and what patterns can be observed across locations?

- Urban vs Rural Comparison

• How do casualty levels differ between urban and rural areas, and which environment is more prone to accidents?

- Environmental Conditions Impact

• How do light conditions (daylight, darkness, street lighting, etc.) affect the number and severity of casualties?

- Seasonal (Monthly) Trends

• What monthly trends are observed in casualty data, and are there any seasonal patterns affecting accident frequency?

- Year-over-Year Trend Analysis

• How have casualties changed over time, and what long-term trends can be identified across different years?

- Combined Risk Factors

• Which combination of factors (road type, vehicle type, and environmental conditions) is most strongly associated with high casualty rates?

PYTHON ANALYSIS QUESTIONS

- • Calculate the total number of casualties in the dataset.

- • Analyze casualties across different severity levels and identify which severity type has the highest impact.

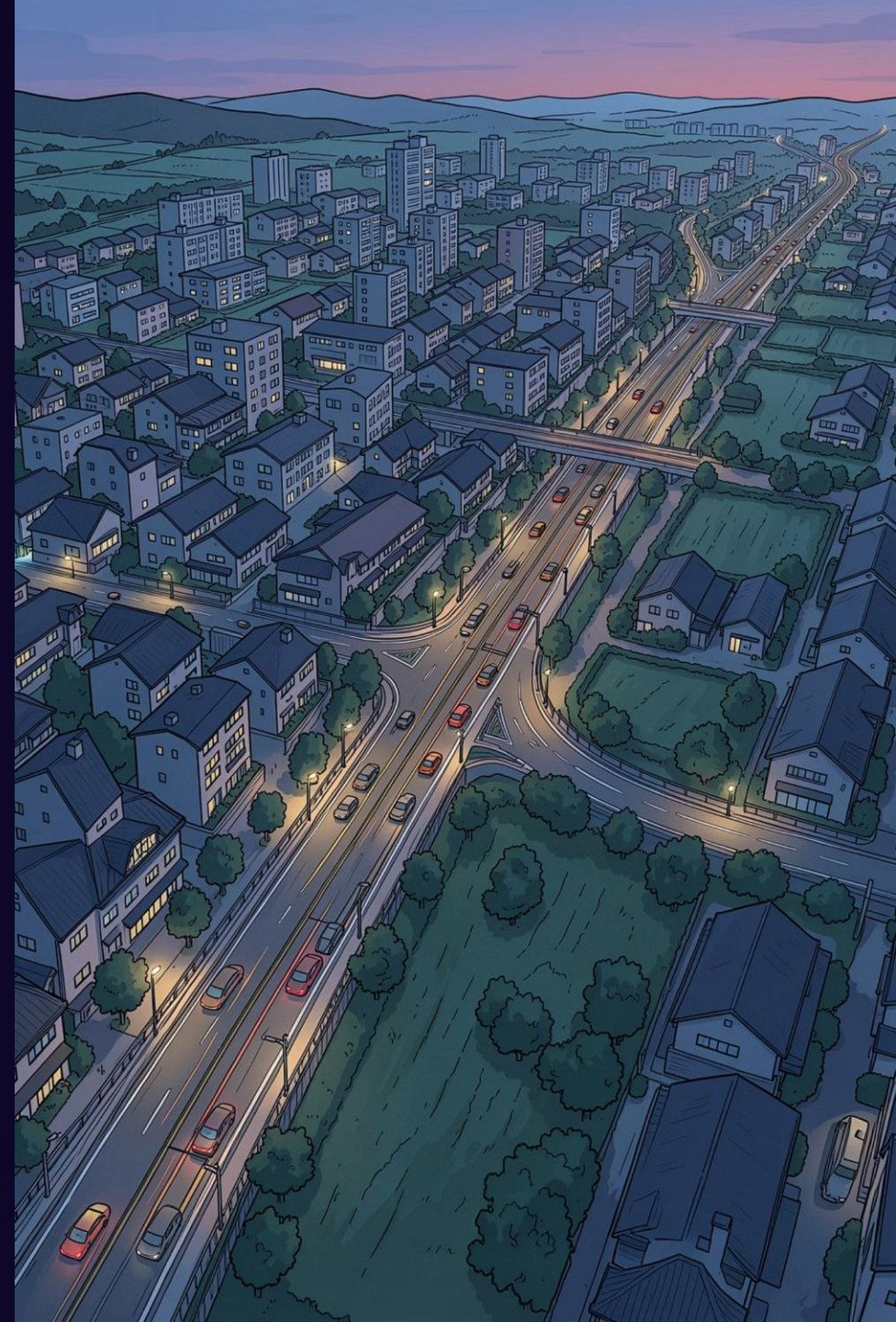
- • Analyze casualties by vehicle type and determine which vehicle type contributes the most to accidents.

- • Analyze casualties by road type and identify the road type with the highest number of casualties.

- • Analyze the distribution of casualties across districts and identify high-risk areas.

Compare casualty levels between urban and rural areas to identify which environment has higher accident risk.

- Analyze how different light conditions affect the number of casualties.
- Analyze month-over-month trends in casualties to identify seasonal patterns.
- Analyze yearly casualty trends to identify increases or decreases over time.
- Perform a multi-factor analysis by grouping road type, vehicle type, and environmental conditions to determine which combinations result in the highest number of casualties.



DASHBOARD OVERVIEW

- The Road Accident Dashboard was developed in Microsoft Excel to provide an interactive visual representation of accident casualty data. The dashboard complements the Python-based analysis by presenting key performance indicators (KPIs), trends, and risk patterns in an easy-to-understand format for decision-makers.



The dashboard includes:



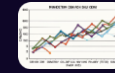
Total casualty count KPI



Casualties by severity level



Vehicle type analysis



Monthly and yearly casualty trends



Casualties by road type



Casualties by road surface condition



Urban versus rural comparison



Day versus night comparison



Interactive filters for year and month selection

- The dashboard enables users to quickly identify high-risk accident conditions, monitor trends over time, and support data-driven road safety planning.

DASHBOARD KEY METRICS

Total Casualties

- Total casualties recorded: 896,568

Severity Distribution

- Slight Severity: 750,646 casualties (84%)
- Serious Severity: 129,439 casualties (14%)
- Fatal Severity: 16,483 casualties (2%)

Vehicle Involvement

- Cars: 695K casualties
- Vans/LGVs: 78K casualties

KPI

TOTAL
CASASUBSIES

KPI

19





- Motorcycles: 76K casualties
 - Buses: 38K casualties
 - Agricultural Vehicles: 3K casualties
 - Other Vehicles: 8K casualties
-

Urban vs Rural Casualties

- Urban Areas: 543K casualties
 - Rural Areas: 354K casualties
-

Day vs Night Casualties

- Daytime: 650K casualties
- Nighttime: 247K casualties

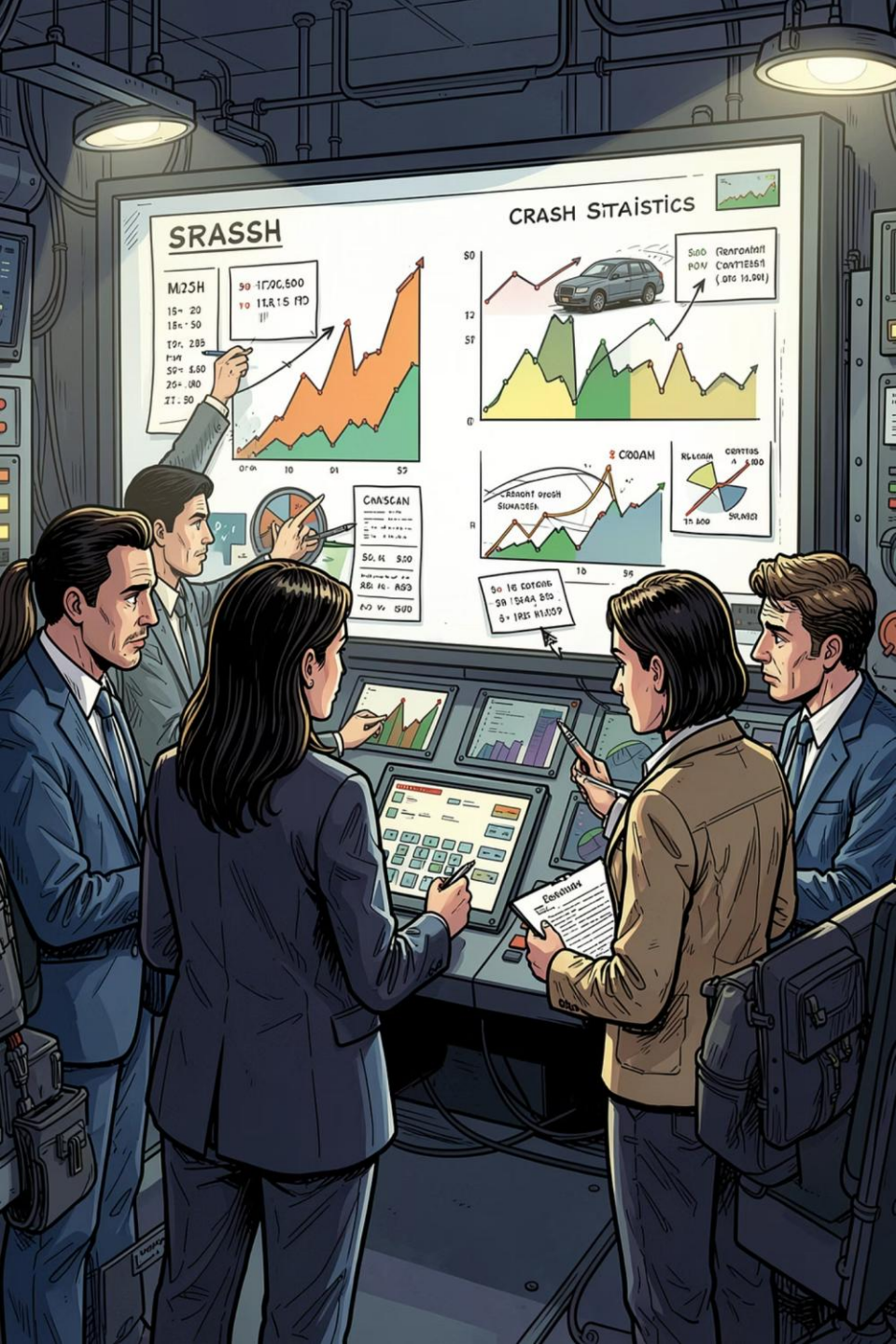
Road Type Distribution

- Single Carriageway: 662K casualties
- Dual Carriageway: 147K casualties
- Roundabout: 56K casualties
- One-Way Street: 16K casualties
- Slip Road: 10K casualties
- Unknown: 6K casualties



Road Surface Conditions

- Dry road surfaces account for the highest number of casualties.
- Wet road surfaces represent the second highest casualty count.
- Snow-covered roads contribute the smallest share of casualties.



DASHBOARD INSIGHTS

Majority of Casualties Are Slight Injuries

Slight injuries account for 84% of all casualties, indicating that while most accidents are not fatal, they still create a significant burden on public safety and healthcare resources.

Cars Are the Most Frequently Involved Vehicle Type

Cars account for approximately 695,000 casualties, significantly exceeding all other vehicle categories. This confirms that private passenger vehicles represent the largest accident exposure group.

Single Carriageway Roads Present the Highest Risk

Single carriageway roads recorded approximately 662,000 casualties, making them the most dangerous road type.

Urban Areas Experience More Casualties

Urban areas account for 543,000 casualties compared with 354,000 in rural areas.



Most Casualties Occur During Daylight Hours

Approximately 650,000 casualties occurred during daytime conditions, compared with 247,000 at night.



Dry Roads Record the Highest Number of Casualties

The majority of casualties occur on dry roads because they are associated with the highest traffic exposure and vehicle usage.



Casualty Levels Show a Declining Trend

The yearly trend chart indicates a gradual reduction in casualties from 2019 through 2022.

Seasonal Patterns and Dashboard Insights



Seasonal Variations Exist Across Months

Monthly casualty trends fluctuate throughout the year, with higher accident counts observed during several peak months.



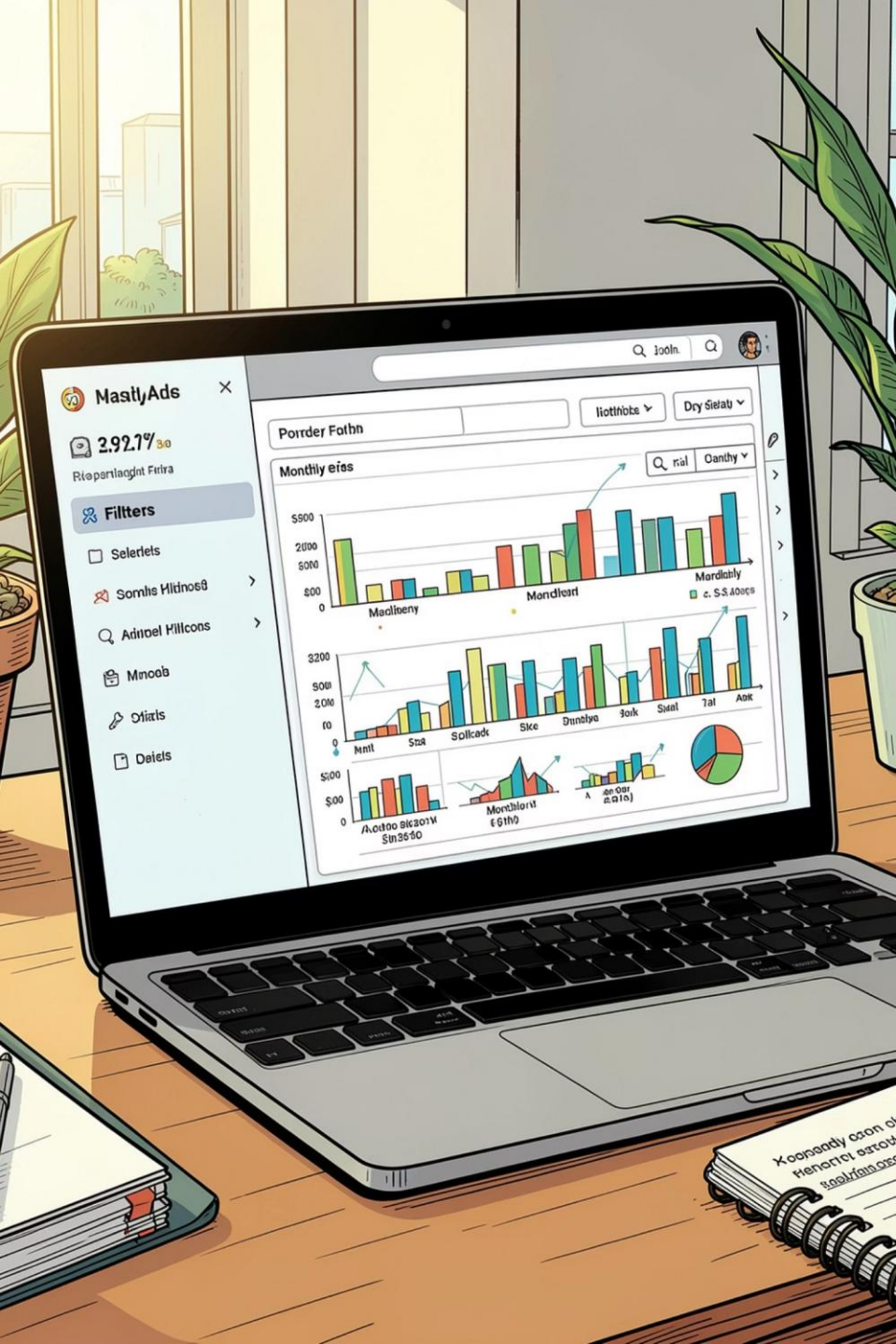
Serious and Fatal Accidents Represent a Smaller Share but Greater Risk

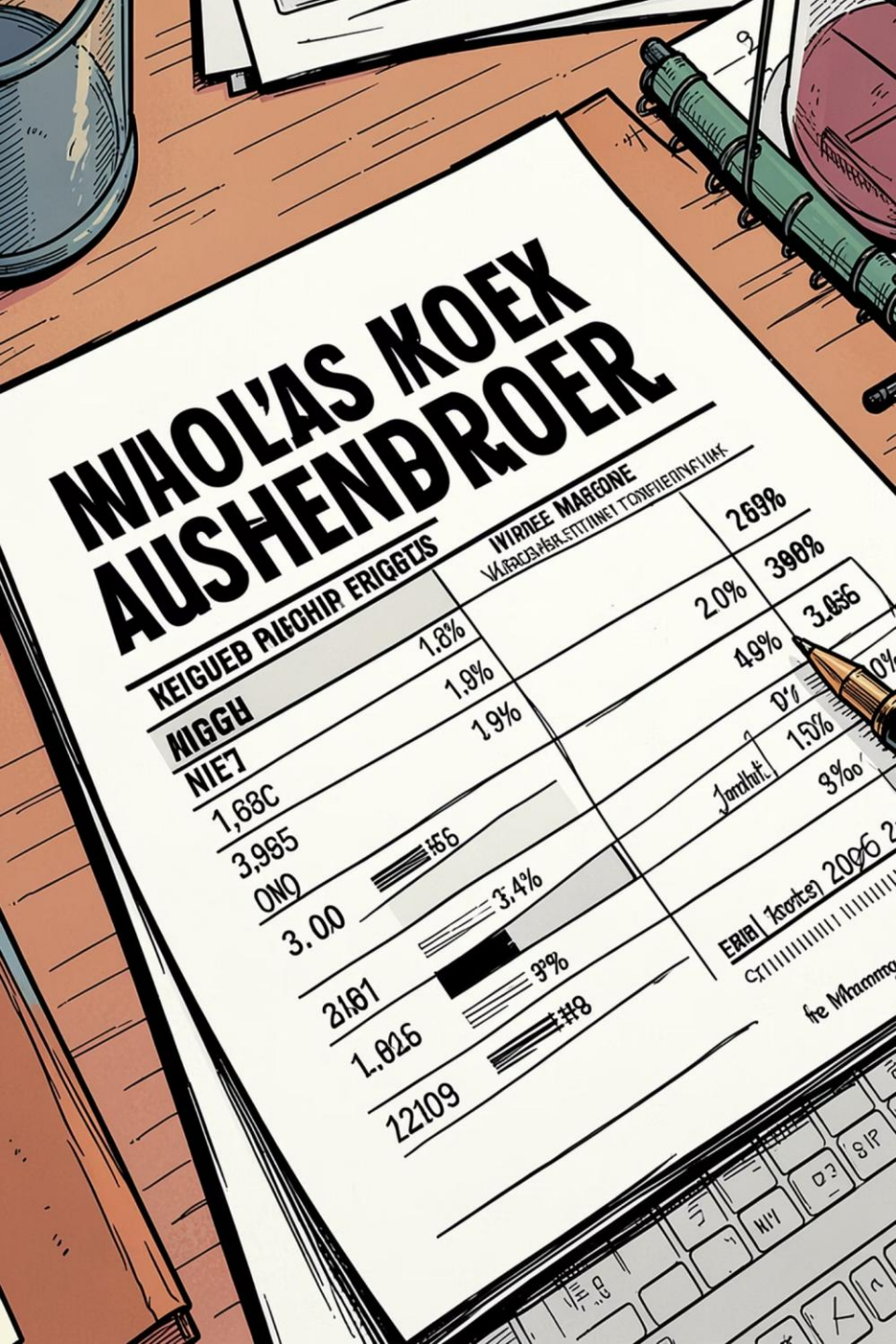
Although fatal and serious accidents account for only 16% of total casualties, they represent the highest public safety concern.



Dashboard Supports Rapid Decision-Making

The interactive filtering capability allows stakeholders to examine casualty patterns by year and month, helping authorities quickly identify trends and prioritize road safety interventions.





KEY INSIGHTS

Overall Impact

• A total of 896,547 casualties were recorded during the analysis period, highlighting the significant impact of road traffic accidents.

Casualties by Severity

• Slight injuries accounted for 750,625 casualties (approximately 84%), making them the most common severity level.

Serious Injuries

• Serious injuries totaled 129,439.

Fatal Accidents

• Fatal accidents accounted for 16,483 casualties.



Casualties by Vehicle & Road Type

- Casualties by Vehicle Type

• Cars were involved in 694,866 casualties, representing the largest share of all vehicle types.

- Casualties by Road Type

• Single carriageway roads accounted for 661,758 casualties, significantly more than any other road type.

- Geographic Risk Analysis

• Birmingham recorded the highest number of casualties among all districts, identifying it as the most critical high-risk location.

- Urban vs Rural Comparison

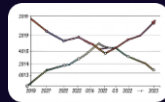
• Urban areas accounted for 543,021 casualties, compared with 353,497 in rural areas.

Light, Yearly Trends, and Combined Risks



Light Conditions

• A total of 649,874 casualties occurred during daylight conditions, compared with 246,673 at night.



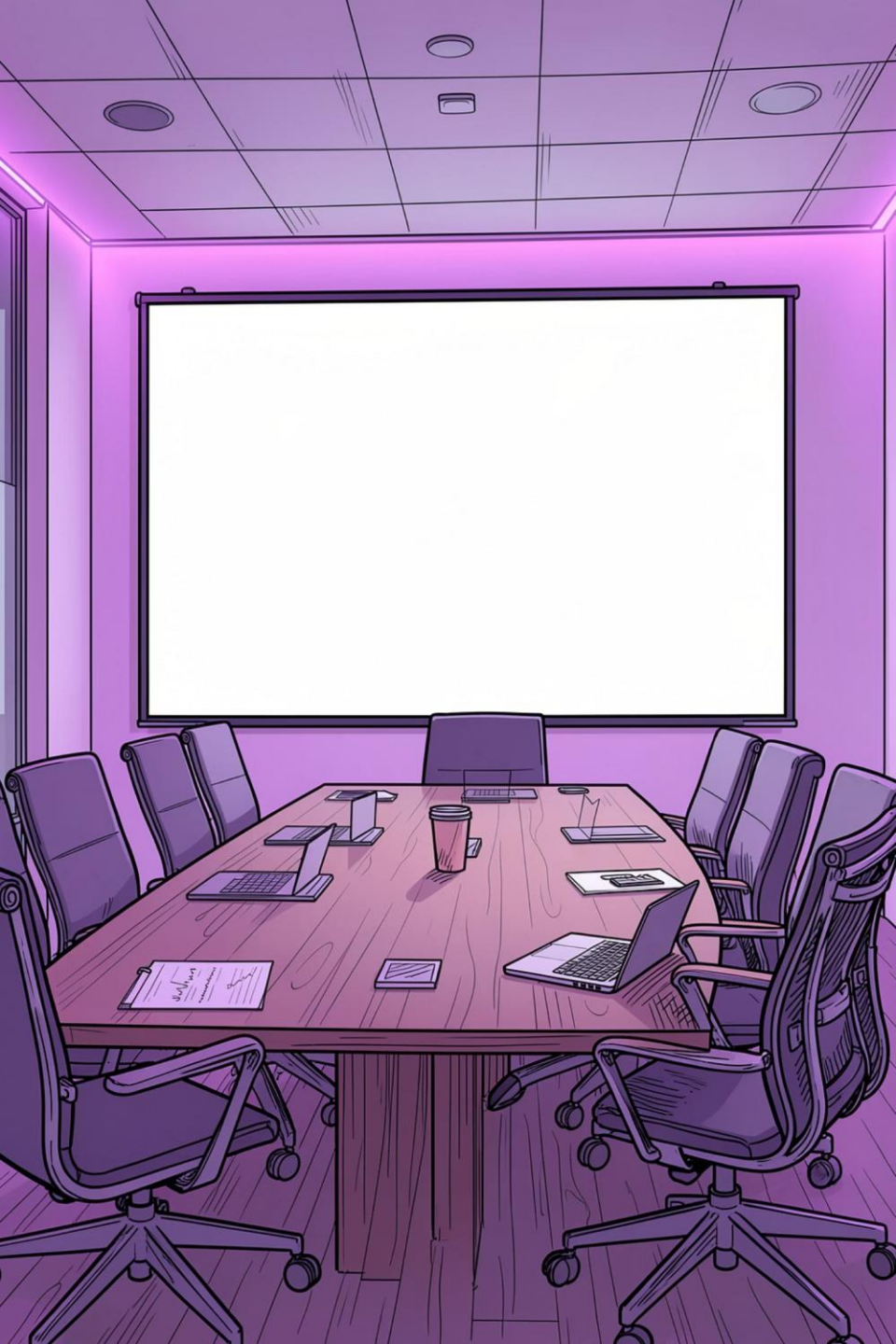
Yearly Trend

• Road casualties declined steadily from 247,776 in 2019 to 195,736 in 2022.



Combined Risk Factors

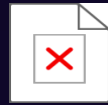
• The highest number of casualties occurred in accidents involving cars on single carriageway roads during daylight conditions.



BUSINESS RECOMMENDATIONS

Strengthen Car Safety Initiatives

- Because cars are associated with the highest number of casualties, authorities should prioritize:



- Speed enforcement



- Seatbelt compliance campaigns



- Driver awareness programs



- Distracted driving prevention

Improve Safety on Single Carriageway Roads

- Given the high casualty rates on single carriageway roads, recommended actions include:



- Installing speed cameras



- Enhancing road signage and lane markings



- Improving lighting and maintenance



- Introducing overtaking restrictions where necessary

Prioritize High-Risk Districts

- Birmingham should be targeted for focused interventions such as:



- Increased traffic police presence



- Hotspot analysis



- Junction redesign



- Pedestrian safety improvements



Enhance Urban Traffic Management

- Urban areas should receive additional investment in:



- Congestion reduction measures



- Traffic signal optimization



- Pedestrian crossings



- Public transport infrastructure

Daytime Enforcement & Seasonal Preparedness



Increase Daytime Enforcement

Sustain Current Safety Programs

Prepare for Seasonal Peaks

Since most casualties occur during daylight hours, enforcement and monitoring should be intensified during peak daytime traffic periods.

The declining trend in casualties suggests that existing road safety measures are effective and should be continued and expanded.

If monthly analysis shows higher casualties in July, October, and November, authorities should deploy seasonal awareness campaigns and additional traffic enforcement during these periods.

Focus on the Highest-Risk Scenario

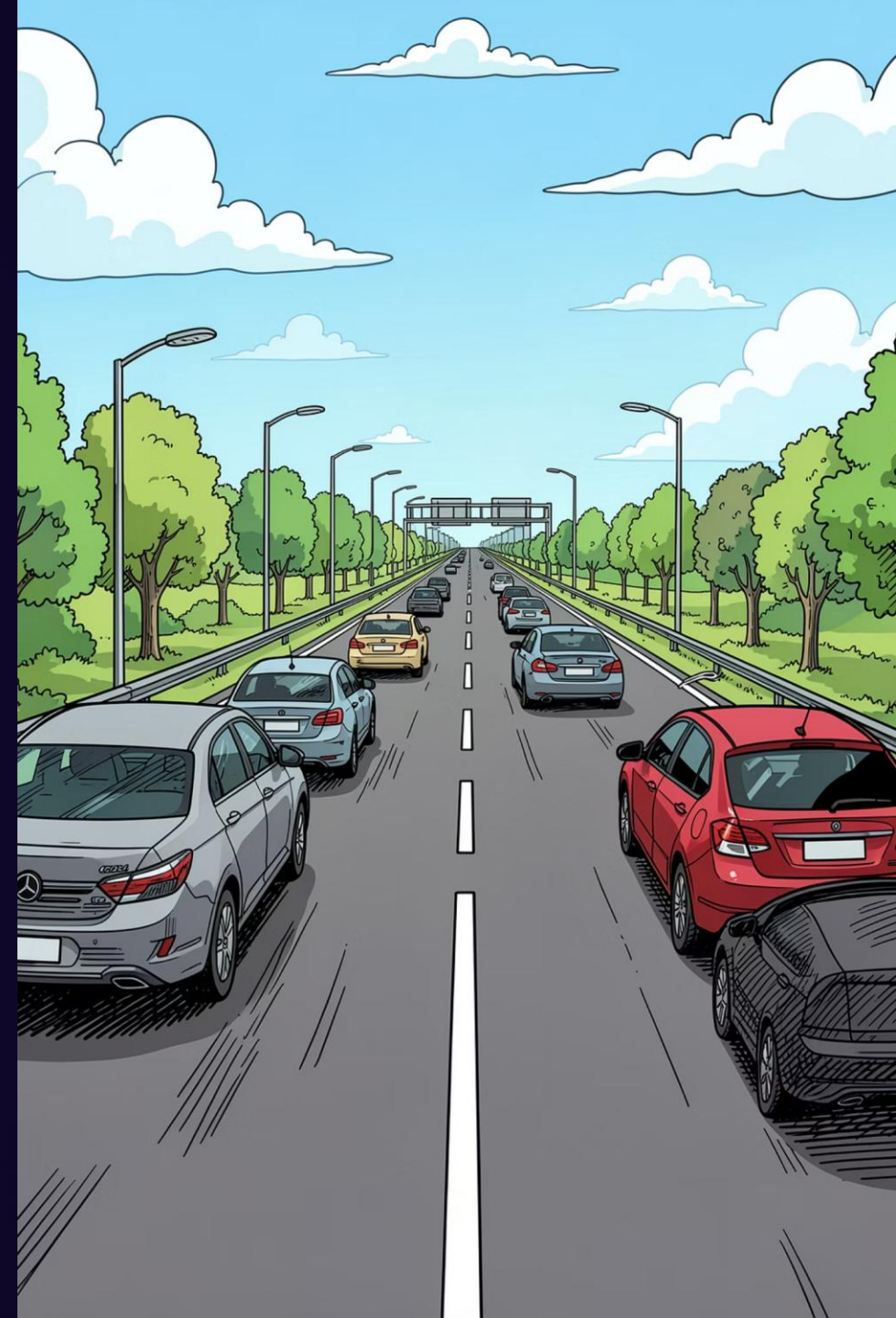
- Target interventions toward accidents involving:

- Cars

- Single carriageway roads

- Daylight conditions

- This combination should be treated as the highest operational priority.



EXECUTIVE SUMMARY

- This project analyzed road accident data to identify the key factors associated with casualty rates and to provide data-driven recommendations for improving road safety.
- A total of 896,547 casualties were recorded during the study period. The analysis showed that:
 - Slight injuries accounted for the majority of casualties.
 - Cars were involved in the highest number of casualties.
 - Single carriageway roads were the most high-risk road type.

DASHBOARD CONCLUSION

- The Excel dashboard successfully transforms complex accident data into an interactive decision-support tool. By visualizing casualty severity, vehicle involvement, road conditions, geographic distribution, and temporal trends, the dashboard provides actionable insights for transportation authorities.



- • Car-related accidents



- • Single carriageway road safety



- • Urban traffic management



Recommendations & Next Steps

- • Daytime traffic enforcement
 - • High-casualty road conditions
 - • Continued implementation of road safety programs
- Together, the Python analysis and Excel dashboard provide a comprehensive framework for understanding accident patterns and supporting data-driven road safety decisions.

